Task 4 decoding after charge adjustment

Nominal-charge sensitivity analysis | WD-007 Satyam Awasthi 13 November 2026

Finding. The pooled adjustment raised accuracy more before stimulation than after it. The waveform-specific correction identified all five waveform classes on its own in the matched cohort.

Adjustment and validation

Each EEG voltage feature was adjusted as $x_{\text{adj}} = x - \hat{\beta} Q_{\text{nom}}$. The frozen Task 3 charge-only slope $\hat{\beta}$ was shared across waveforms or selected by waveform. Nominal charge Q_{nom} used the known waveform area and an amplitude lookup inferred from the Task 4 CSV's second presentation. The same correction vector was applied to the matched pre-stimulus window.

The analysis reused the WD-001 comparison-held-out folds, 14 electrodes, nine 50 ms bins and shrinkage LDA. The original pre- and post-stimulus predictions were reproduced. The pooled model used 1,244 trials from all ten participants. The direct comparison of both slopes used 854 trials from seven participants with all five waveform calibrations in the original strict Task 3 fit.

Held-out waveform decoding

Cohort	EEG window	Original	Pooled slope	Waveform-specific slope
All ten	Post	19.05%	26.35%	—
All ten	Pre	16.75%	26.12%	—
Matched seven	Post	20.73%	28.09%	99.87%
Matched seven	Pre	16.73%	26.62%	99.77%

Scores are participant-mean balanced accuracy. The five-class chance reference is 20%.

The pooled gain was 7.30 percentage points post-stimulus and 9.37 points pre-stimulus. The paired difference in gains was −2.07 points (post minus pre, 95% bootstrap interval −4.46 to +0.05).

Controls without EEG

LDA scored 20% on nominal charge alone and on each correction vector alone. With one fixed template per participant and waveform, these controls have zero within-class covariance. On the same folds, nearest-centroid decoding reached 76.00% on both nominal charge and the pooled correction across all ten participants. Charge overlap accounts for that limit. The waveform-specific correction alone reached 100.00% in the matched seven because selecting the slope used waveform identity.

The pre-stimulus gains and template controls show that the adjustment itself can introduce waveform information. A clean Task 4 test would require trial-level charge measured independently of the waveform labels.

Analysis record on GitHub | Interactive WD-007 report on HTMLPreview.